Web Development

HTML

❖ HTML BASIC TAGS

These tags are fundamental building blocks for creating web documents.

1. <!DOCTYPE html>

- Specifies the document type and version of HTML being used.
- Syntax → <!DOCTYPE.html> (HTML5 version>

2. <html>

- The root element that contains all other elements on the page.It includes the <head> and <body> sections.
- Syntax →

```
<html>
Content write here
</html>
```

3. <head>

- Contains metadata about the document, such as character encoding, title and links to external resources like stylesheets and scripts.

HIRA KUMAR

- Syntax → <head>
  Metadata and links go here
  </head>

4. <title>

- It contains the title of the whole HTML page, and this name is displayed in the browser's title bar or tab.
- Syntax → <title>
  Your Web page name/title
  </title>

5. <meta>

- Specifies metadata about the document,such as character encoding and viewport settings.
- Syntax → <meta charset="UTF-8">

6. <body>

- It contains the visible content of the web page, including text, images, links, and more.
- Syntax → <body>
  Content written here will be displayed in the browser
  </body>

7. <h1>,<h2>,<h3>,<h4>,<h5>,<h6>

- Headings of different levels, used to structure content with <h1> being the highest level and <h6> the lowest.
- Syntax → <h1> Main heading </h1>

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Heading check</title>
    </head>
    <body>
        <h1> Hello World ! </h1>
        <h2> Hello world ! </h2>
        <h3> Hello world !</h3>
        <h4> Hello world !</h4>
        <h5> Hello world !</h5>
        <h6> Hello world !</h6>
    </body>
</html>
```

Hello World !

Hello world !

Hello world !

Hello world !

Hello world !

Hello world !

8.

- Defines a paragraph of text.
- Syntax → This is a paragraph

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Heading check</title>
    </head>
    <body>
        <p> <h1> Hello, I am hira ! </h1></p>
        <p> May i help you ?</p>
    </body>
</html>
```

Hello, I am hira !

May i help you ?

9. Entity

- In HTML, an entity is a special code used to display reserved characters or symbols that cannot be typed directly in normal HTML.

➢ **Why HTML Entities are Used**

Some characters have special meaning in HTML, like:

- < (used for tags)
- > (used for tags)
- & (used to start an entity)

So we use **entities** to display them correctly on the webpage.

Symbol	Entity Name	Entity number
<	<	<
>	>	>
&	&	&
“	"	"
Space		

More symbol search online…. (math symbol, emoji, currency,……)
preferred website/link →

1. https://www.w3schools.com/html/html_symbols.asp
2. https://www.toptal.com/designers/htmlarrows/symbols/

Note

  (space – 4 space contain)
  (space – 1 space contain)
 (space – 2 space contain)

Home work

1. Create an HTML page to display a heading and a paragraph about yourself (e.g. name, class, roll no, mobile number,…..) on the browser.
2. Create a simple HTML page using only basic tags to display a heading and two paragraphs about your favorite subject.

❖ HTML formatting tags

- HTML offers an array of formatting tags, empowering you to regulate the visual styling of text and content within a web page.

1. <b>

- Mark text bold
- Syntax → any text

****

✓ Represents Strong importance and is typically rendered as bold text.

✓ Syntax → any text

2.

- Renders text in italics.
- Syntax → any text

****

✓ Represents emphasized text and is typically rendered in italics.

✓ Syntac → any text

3. <u>

- Underlines text
- Syntax → <u> any text </u>

4. <s> or <strike>

- Renders text with a strikethrough.
- Syntax → <s> any text </s> **OR**
  <strike> any text </strike>

5. <sup>

- Makes text superscript (e.g. – a^2+b^2)
- Syntac → ^{any text}

6. <sub>

- Makes text subscript (e.g – H_2O)
- Syntax → _{any text}

7. <Hr>

- Used to display a horizontal ruler, used to separate content.
- Syntax → <hr> (not close tag)

```html
<!DOCTYPE html>
<html>
    <head>
        <title> hira </title>
    </head>
<body>
<p> Hello Wolrd !</p> <hr>
<p> Hello India !</p> <hr> <hr>
<p> Hello Bihar ! </p><hr>
</body>
</html>
```

Hello Wolrd !

Hello India !

Hello Bihar !

---

8. <mark>

- Used to text highlight color
- Syntax → <mark> any text </mark>

```html
<p> hello, i am <mark>hira kumar</mark>.</p>
```

hello, i am hira kumar.

9. <bdo>

- Bdo means Bi-directional override.
- It it used change text direction right to left(rtl) or left to right(ltr)
- Syntax → <bdo dir="rtl"> any text </bdo>

```html
<h2><bdo dir="rtl"> HIRA </bdo></h2>
```

ARIH

10. <address>

- Used for address/ contact details in html page.
- Syntax <address> any address </address>

```html
<address>
    Devi asthan, hanuman gali <br>
    Biharsharif, Nalanda <br>
    803101
</address>
```

Devi asthan, hanuman gali
Biharsharif, Nalanda
803101

Font Style and Size

(Not recommended, use CSS for styling instead)

11. <font> (not recommended)

- Sets font size, color, and face (font name).
- Syntax → <font size="4" color="red" face="arial"> any text </font>

12. <big>

- The <big> tag was used to make text **one size larger** than the surrounding/normal text.
- Syntax → <big> any text </big>

13. <small>

- The <small> tag was used to make text **one size smaller** than the surrounding/normal text.
- Syntax → <small> any text </small>

index.html > html > body > p

```html
<!DOCTYPE html>
<html>
    <head>
        <title> hira </title>
    </head>
<body>
<p> hello, may i help you ? </p> <hr>
<big> hello, may i help you ?</big> <hr> <hr>
<small>hello, may i help you ? </small><hr>
</body>
</html>
```

hello, may i help you ?

hello, may i help you ?

hello, may i help you ?

Text alignment

14. {CSS} (division/container)

- Defines a division or section for styling purpose and can be used to control text alignment.
- "div" is act as container, mostly use create layout design and CSS.

```
<div>
  <h2>Heading</h2>
  <p>This is inside div.</p>
</div>
```

- Align is a attribute.
- It is written inside the opening tab.
- Syntax → any text {CSS}
  OR
   any text {html}
  OR
  <h1 align="center"> any text </h1> {html}
  OR
   any text {html}

Line Breaks and Whitespace

15.

- Inserts a line break within text (new line).
- Syntax →
 (not close this tag)

```
nl > html > body > p
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Heading check</title>
    </head>
    <body>
        <h2>
        <p> <b> Hello hira </b> </p>
        <p>  Again, <strong>Hello hira</strong></p>
        <p> <i>May i help you ?</i></p>
        <p> i live in <u>india</u></p>
        <p> Hello <s> World</s> India</p>
         </h2>
        <p> <font size ="5" color="red" face="arial"> (a-b)<sup>2</sup> =(a+b)(a-b) </font></p>
        <p> <font size ="6" color="green" face="arial"> Water formula is = H<sub>2</sub>O</font></p>
        <div style ="text-align:center;"><h1>The End</h1></div>
        <p> Three input device <br> Keyword <br> Mouse <br> Camera</p>
         <p> Q. What is Your name ? <br> My name is Hira kumar</p>
         <p> Q. What is Your name ? <pre> My name is Hira kumar</pre></p>
     </body>
</html>
```

Output

Hello hira

Again, Hello hira

May i help you ?

i live in <u>india</u>

Hello ~~World~~ India

$(a-b)^2 = (a+b)(a-b)$

Water formula is = H_2O

The End

Three input device
Keyword
Mouse
Camera

Q. What is Your name ?
My name is Hira kumar

Q. What is Your name ?

```
My name is Hira kumar
```

16. <pre>

- Performatted text as it is without ignoring spaces & next line.
- Syntax → <pre> any text </pre>

index.html > html > body > pre

```
<!DOCTYPE html>
<html>
    <head>
        <title> hira </title>
    </head>
<body>
<p> Computer is an                    machine ? </p>
<pre> Computer is an                  machine ?
Collection of program is called             ?
</pre>
</body>
</html>
```

Computer is an machine ?

```
 Computer is an                  machine ?
Collection of program is called            ?
```

17. <q>

- Defines a short inline quotation.

- Syntax → <q> any text </q>

18. <abbr>

- Represents an abbreviation (संक्षिप्त) or acronym, and you can provide a title attribute for the full text.
- Syntax → <p><abbr> any text </abbr><p>

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Heading check</title>
    </head>
    <body>
        <p> My name is <q>Hira Kumar</q></p>
        <p><abbr title ="Bachelor of Computer Application">BCA</abbr> is amazing !</p>

    </body>
</html>
```

My name is “Hira Kumar”.

BCA is amazing !

Home Work

1. Write an HTML program using basic tags to show your name in bold, your city in italic, and align the text to the right.
2. Create an HTML page to display a title, a centered heading, and a paragraph containing bold,italic and underlined text.
3. Write a Simple HTML program (not use JS) to display 1 to 10 counting in horizontal and vertical way.